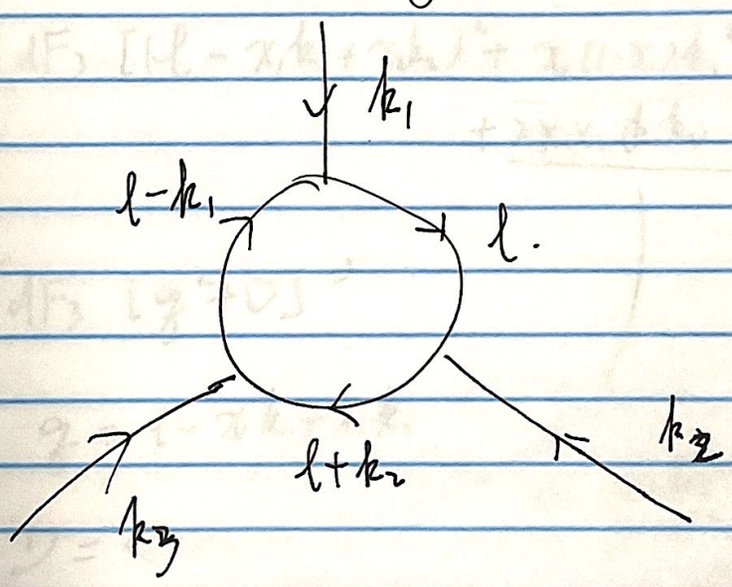
Roadmap. Srednicki §16, start of §17: the last divergent one-loop function. The vertex V_3 in $d = 6 - \epsilon$ fixes $Z_g = 1 + C$ up to a finite κ_C, a convention ($V_3(0,0,0) = g$); at large momenta V_3 runs like $\ln k^2$. The box V_4 is finite. Next: power counting and $2 \to 2$ (LecNote14, §§18–20); schemes and the RG (LecNote15, §§26–28). Supplements: FeynmanRulesLSZ, DimRegMSbar, DimensionalAnalysis_hbar, RenormalizationScaleGroup, RunningCoupling, OneLoopHighEnergy, iPiassignment.

Other Loops & Renormalizability.

Z_φ, Z_m, Γ, Z_g.

$$i\mathbf{V}_3(k_1,k_2,k_3) = iZ_g g + (-ig)^3 \left(\tfrac{1}{2}\right)^3 \int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}((l-k_1)^2)\tilde{\Delta}((k_2+l)^2)\tilde{\Delta}(l^2) + \mathcal{O}(g^5).$$



This time, I can't do geometric series for all ~~orders~~.

Let's do it order by order.

$$\Delta((l-k_1)^2)\,\Delta((k_2+l)^2)\,\Delta(l^2)$$

$$= \int dF_3\ [x_1(l-k_1)^2 + x_2(l+k_2)^2 + x_3 l^2 + m^2]^{-3}$$

$$\int dF_3 = 2\int_0^1 dx_1\,dx_2\,dx_3\ \delta(x_1+x_2+x_3-1)$$

$$= \int dF_3\ [(l - x_1k_1 + x_2k_2)^2 + x_1(1-x_1)k_1^2 + x_2(1-x_2)k_2^2 + 2x_1x_2(k_1\cdot k_2) + m^2]^{-3}$$

$$= \int dF_3\ [q^2 + D]^{-3}$$

$$q = l - x_1k_1 + x_2k_2$$

$D =$ ← using that $k_3 = -(k_1+k_2)$.

$$= x_3x_1k_1^2 + x_3x_2k_2^2 + x_1x_2k_3^2 + m^2.$$

With Rotation $q \to \bar{q}$.

$$\Gamma^{(2)}(k_1,k_2,k_3)/g = Z_g + g^2\int dF_3 \int \frac{d^d\bar{q}}{(2\pi)^d}\,\frac{1}{(q^2+D)^3} + O(g^4)$$

It diverges for $d \geq 6$

[1] Here μ is LecNote12's $\tilde{\mu}$: $g \to g\tilde{\mu}^{\epsilon/2}$, so g^2 brings $(\tilde{\mu}^2)^{\epsilon/2}$ and the pole is $+\alpha/\epsilon$. Only $C \ni -\alpha/\epsilon$ is forced (all of MS-bar's C, LecNote15); $-\alpha \ln(\mu/m)$ trades μ for m, and κ_C is free (p. 4). The rest needs no regulator: in $d = 6$, $V_3(k) - V_3(0) = -\frac{1}{2}\alpha g \int dF_3 \ln(D/m^2)$.

$$\int \frac{d^d q}{(2\pi)^d} \frac{1}{(q+D)^3} = \frac{\Gamma(3-\frac{1}{2}d)}{2(4\pi)^{d/2}} D^{-(3-d/2)}$$

for $\epsilon > 0$, it converges.

$\Rightarrow$ for $d = 6-\epsilon$, $g \to g\mu^{\epsilon/2}$ (1)

$$V_3(k_1,k_2,k_3)/g = Z_g + \frac{1}{2}\alpha\,\Gamma(\tfrac{\epsilon}{2}) \int dF_3 \left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\epsilon/2} + O(\alpha^2)$$

$\Gamma(\frac{\epsilon}{2}) = \frac{2}{\epsilon} - \gamma + O(\epsilon^2)$

$A^{\epsilon/2} \simeq 1 + \frac{\epsilon}{2}\ln A + O(\epsilon^2)$

Taking the $\epsilon \to 0$ limit.

$$= Z_g + \frac{1}{2}\alpha\left[\frac{2}{\epsilon} + \int dF_3 \ln\left(\frac{4\pi\tilde{\mu}^2}{e^{\gamma} D}\right)\right] + O(\alpha^2)$$

Knowing $\int dF_n = 1$, letting $\mu^2 = 4\pi e^{-\gamma}\tilde{\mu}^2$,

setting $Z_g = 1 + C$, we get

$\ln \frac{\mu^2}{D} = 2\ln\frac{\mu}{m} - \ln\frac{D}{m^2}$

$\int dF_3 \ln\frac{\mu}{m} = \ln\frac{\mu}{m}$.

Consider to switch the notation m_0 to be Λ or μ_0. μ_0 was used before. Let's use μ_0. But for now use m for us to introduce IR divergence. Then

$$= 1 + \left\{\alpha\left[\frac{1}{\epsilon} + \ln(\mu/m)\right] + C\right\} - \frac{1}{2}\alpha\int dF_3 \ln(D/m^2) + O(\alpha^2)$$

Let's choose $C = -\alpha\left[\frac{1}{\epsilon} + \ln(\mu/m) + \kappa_C\right] + O(\alpha^2)$

$$V_3(k_1,k_2,k_3)/g = 1 - \frac{1}{2}\alpha\int dF_3 \ln(D/m^2) - \kappa_C\alpha + O(\alpha^2)$$

1 Why none: LSZ fixes only m and the residue (FeynmanRulesLSZ), and no on-shell point exists ($k_1 \cdot k_2 = m^2/2$ needed; real on-shell momenta have $|k_1 \cdot k_2| \geq m^2$). Another κ_C only redefines g by $1 + O(\alpha)$, which drops out once g is fixed by a measurement. MS-bar (LecNote15): $\kappa_C = -\ln(\mu/m)$.

Sounds arbitrary, isn't it? ④

There is no other B.C. to help me fix κ_c. ①

so (now a choice "scheme") can be made. we'll return to this

for instance, we can set $\kappa_c = 0$, then $+O(\alpha^2)$.

$$V_3(0,0,0)/g = 1 - \frac{1}{2}\alpha \int dF_3 \ln(D/m^2) + O(\alpha^2)$$

$$= 1.$$

This time, the $\int dF_3 \ln(D/m^2)$ cannot be done in a closed form, but it is clear for $|k_i^2| \gg m^2$

2 The $O(1)$: for one large k_1^2 (bottom) the dropped $\int dF_3(\ln x_1 + \ln x_3) = -3$, so $V_3/g \simeq 1 - \frac{1}{2}\alpha[\ln(k_1^2/m^2) - 3]$. LecNote14 p. 8 uses it with $-s$ as the large invariant: $\ln(-s/m^2) = \ln(s/m^2) - i\pi$ from $D - i\epsilon$ (iPiassignment).

$$V(k_1, k_2, k_3)/g \simeq 1 - \frac{1}{2}\alpha[\ln(k^2/m^2) + O(1)] + O(\alpha^2)$$ ②

3 Not yet the coupling: each vertex also carries three half-propagators. At $k_i^2 = k^2 \gg m^2$, with the margin's Π, $V_3[k^2\tilde{\Delta}]^{3/2} \simeq g[1 - (\frac{1}{2} - \frac{3}{2} \cdot \frac{1}{12})\alpha \ln \frac{k^2}{m^2}] = g[1 - \frac{3}{4}\alpha \ln \frac{k}{m}]$: weaker at high k, asymptotic freedom. The same $-\frac{3}{4}\alpha/\epsilon$ is the pole of $C - \frac{3}{2}A$; LecNote15: $d\alpha/d\ln\mu = -\frac{3}{2}\alpha^2$.

vertex strength is changing ③ with external momentum

Recall $D = x_1x_3k_1^2 + x_2x_3k_2^2 + x_1x_2k_3^2 + m^2$

Let $k_1^2 \gg k_2^2, k_3^2, m^2$.

$D \simeq x_1x_3k_1^2$

$$\int dF_3 \ln(D/m^2) \simeq 2\int dx_1 dx_2 dx_3 \ln(x_1x_3k_1^2/m^2)\delta(1 - x_1 - x_2 - x_3)$$

$$= 2\int_0^1 dx_1 \int_0^{1-x_1} dx_3 \ln(x_1x_3k_1^2/m^2)$$

$$\simeq \int dF_3 (\ln x_1 + \ln x_3) + \ln(k_1^2/m^2))$$

$$\simeq \int dF_3 \ln(k_1^2/m^2)$$

$$= \ln(k_1^2/m^2).$$

Recall is the propagator case, we have

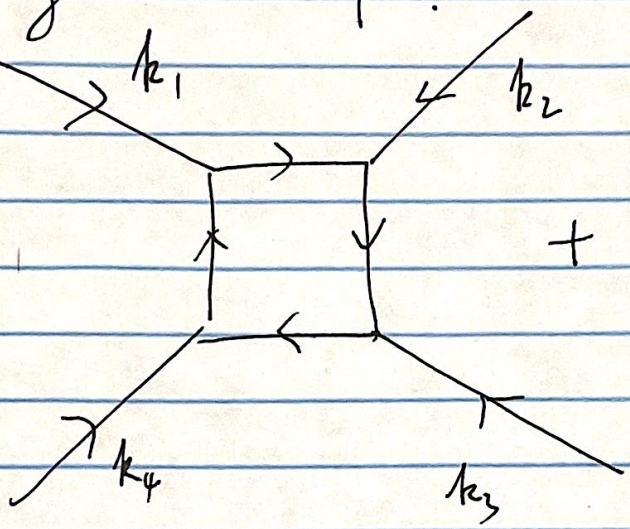
$$\Pi(k^2) = \frac{1}{12}\alpha[C_1 k^2 + C_2 m^2 + 2k^2 f(r)] + O(\alpha^2)$$

In the large $k^2 \gg m^2$ limit

$$\frac{\Pi(k^2)}{k^2 + m^2} \simeq \frac{1}{12}\alpha[C_1 + \ln\frac{k^2}{m^2}]$$

A common feature is $[O(1) + \ln\frac{k^2}{m^2}]$ renormalization over the original (while Z_3 is absorb the $\frac{1}{\epsilon}$ parts)

Any more 1-loops?



$+ (k_3 \leftrightarrow k_2) + (k_3 \leftrightarrow k_4)$.

$$i\mathcal{V}_4 = g^4 \int \frac{d^d l}{(2\pi)^d} \quad \Delta \; \Delta \; \Delta \; \Delta$$

$+ (k_3 \leftrightarrow k_2) + (k_3 \leftrightarrow k_4)$.

$$\mathcal{V}_4 = \frac{g^4}{6(4\pi)^3} \int dF_4 \left(\frac{1}{D_{1234}} + \frac{1}{D_{1324}} + \frac{1}{D_{1243}} \right) + O(g^6).$$

[1]

$\frac{1}{(q^2+D_{1234})^4}$.

For $d=6$

$$\int \frac{d^6 l}{(2\pi)^6} \frac{1}{(q^2+D)^4} = \frac{i}{6(4\pi)^3 D}.$$

Finite

1 D_{1234}: put $x_1, \ldots, x_4$ on the lines between legs 4&1, 1&2, 2&3, 3&4; on shell $D_{1234} = [1-(x_1+x_3)(x_2+x_4)]m^2 - x_1x_3s_{12} - x_2x_4s_{23}$, $s_{ij} = -(k_i+k_j)^2$ (LecNote14 p. 7's D_4). Finite: $\omega = 6 - 2\cdot 4 = -2$, and every one-loop $E \geq 4$ function has $\omega = 6 - 2E < 0$. "See next": higher loops, by power counting (LecNote14, §18).

We've done it!

(Or, have we done it?)

(See next)